

Proprioception Drift in Virtual Reality: An Experiment with an Unrealistically Long Leg

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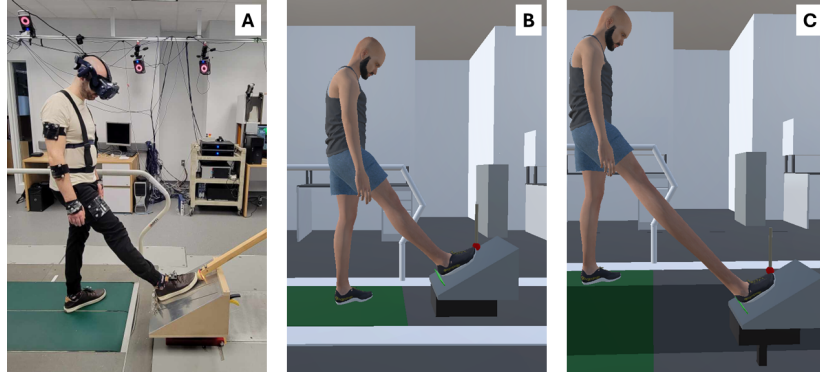


Figure 1: (A) Participant receiving visuotactile stimuli, (B) with normal virtual legs, and (C) with one lengthened virtual leg.

ABSTRACT

Embodiment refers to the sensation of owning, controlling, and perceiving a virtual or artificial body as one's own. This study investigates how embodying an avatar with a leg twice its normal length affects proprioception, with congruent or incongruent visuotactile stimuli. Preliminary results ($n = 10$) show that participants experienced embodiment with a lengthened virtual leg, regardless of congruence of stimuli. A proprioceptive drift of 31.2 cm toward the virtual foot was observed. These findings extend research on upper-body proprioception to include virtual lower-limb deformations.

Index Terms: Self-Avatar, Sense Of Embodiment, Proprioceptive Drift

1 INTRODUCTION

Proprioception provides sensory information about the spatial orientation of body segments in relation to each other and the surrounding environment, relying on input from muscles, tendons, and joint receptors. It influences posture, movement awareness, and coordination and works alongside sensory systems like vision and tactile feedback to shape body perception and self-awareness [1]. Immersive virtual reality (VR) allows the study of how external body parts can be perceived as one's own, a phenomenon referred to as the sense of embodiment. This arises from the integration of visual, tactile, and proprioceptive inputs, fostering a sense of ownership over an artificial limb or external objects [2]. Misalignment between virtual and real limb positions can cause a proprioceptive drift, where the perceived position of the limb shifts toward the virtual representation. This shift reflects the brain's integration of proprioceptive and visual information to form an updated body repre-

sensation [3]. Proprioceptive drift in response to misalignment has been extensively studied in upper limbs. Kilteni et al. [4] further demonstrated that proprioceptive drift occurs even when a virtual arm is elongated to four times its original size. Participants maintained a sense of embodiment and experienced a drift proportional to the arm's deformation. In lower limbs, this phenomenon has been much less explored, although Flögel et al. [5] and Crea et al. [6] demonstrated drift toward a rubber foot in seated participants.

To our knowledge, no research has yet examined proprioceptive drift in virtual lower-limb deformations, despite several reasons suggesting it may differ from observations in the upper limb. For example, lower limbs are primarily involved in weight-bearing, locomotion, and postural balance. Their control relies less on visual input and is supported by a smaller cortical representation compared to the upper limbs. To address this gap, we adapted Kilteni et al.'s approach to the lower body by lengthening one virtual leg.

2 MATERIALS AND METHODS

The experiment used a Vicon motion capture system (12 cameras, 200 Hz) to track 16 rigid bodies with reflective markers, enabling real-time mapping of participants' movements to a virtual avatar customized to match the participant's height, rendered in MakeHuman 1.1.0 and Unity (v2022.3.26). Participants wore an HTC Vive Pro Eye HMD for a first-person view. A custom footrest allowed consistent leg positioning - leg fully extended and 30° ankle flexion (Figure 2). Virtual leg deformities were created by scaling the shank segment to twice its normal length, with gradual changes applied over 10 seconds. A wooden stick with a foam ball attached was utilized to provide visuotactile stimuli. The stick featured three reflective markers, allowing its movements to be tracked in real time via motion capture cameras, to render a virtual stick in the virtual environment.

Ten healthy volunteers [6 females, 4 males; age ($M = 29.1$, $SD = 4.1$ years); height ($M = 170.1$, $SD = 9$ cm)] participated in the study. The 2x2x2 within-subject design included three factors: leg deformation (N:normal or L:lengthened), limb dominance (D:dominant or nD:non-dominant), and visuotactile stimuli (C:congruent or I:incongruent). In the incongruent condition, the virtual stick was offset by 0.15 m anterior to the feet (Figure 2). At

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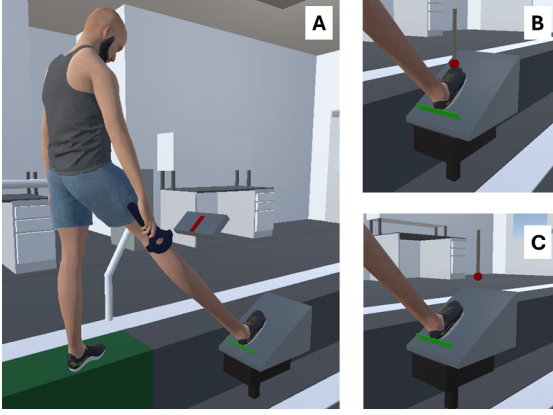


Figure 2: (A) Participant choosing the indicator's position. (B) Congruent and (C) Incongruent visuotactile stimuli.

the beginning of each condition, presented in pseudo-random order, participants put on an HMD and viewed their virtual body from a 1PP, in a virtual replica of the laboratory. They performed arm and leg circumductions, stepped backward, and tapped their feet on the floor while observing themselves in a virtual mirror for one minute. Afterward, the footrest appeared in the VE, with a position that varied depending on the studied limb and the length of the virtual limb. Participants placed their foot on it with their leg fully extended. Visuotactile stimuli were applied by gently stroking the top of the foot twenty times (Figure 1). Next, an indicator (Figure 2) appeared randomly between the virtual knee and toes. Participants used an HTC controller trackpad to slide the indicator along the virtual leg's vertical axis, aligning its red line with their perceived toe position. This was repeated three times. Foot position estimates were calculated as the mean distance of the indicator from the virtual knee. Finally, embodiment was assessed via a validated questionnaire by Peck et al. [7], and participants had to walk around the room to avoid any habituation to the previous self-avatar.

3 RESULTS

Table 1 shows means, standard deviations, and 3-way ANOVA main effects for *Estimated foot position* and *Embodiment*. No interaction effects were significant. Pairwise comparisons (Bonferroni-Holm corrected) showed that participants with a lengthened leg placed the indicator significantly farther ($p < .001$) than those with normal legs. By regrouping data based on the *Deformation* factor, the mean proprioceptive drift was 31.2 cm. Embodiment was higher in *NDC* than all *Incongruent* stimuli ($p \leq .002$), and *NDI* had lower embodiment than *NnDC* and *LDC* ($p \leq .002$).

4 DISCUSSION AND CONCLUSION

Preliminary results indicate that lengthening one limb did not alter the level of embodiment with a self-avatar, regardless of leg dominance. However, embodiment increased with visuotactile congruence, consistent with Kokkinara's findings [1]. Adding more participants could further highlight the differences between congruent and incongruent conditions. Nonetheless, all stimuli were temporally synchronized with participants' movements, a factor known to preserve embodiment [8]. Studies aiming to reduce embodiment might consider applying asynchronous stimuli, as demonstrated in Kokkinara's study, which observed lower embodiment when a virtual ball randomly tapped participants' legs, irrespective of tracking position [1]. Regarding proprioception, lengthening one leg consistently induced the illusion that the embodied foot was farther away, resulting in a mean proprioceptive drift of 31.2 cm across deformation conditions. The mean lengthening of the virtual shank was

Table 1: Means, SDs, and 3-way ANOVA results. **D/nD**: dominant/non-dominant leg; **C/I**: congruent/incongruent stimuli.

Condition	Est. foot pos. (cm)	Embod. score
Normal leg		
DC	50.4 (2.9)	4.81 (0.71)
DI	48.3 (4.5)	4.01 (0.76)
nDC	50.4 (3.2)	4.56 (0.82)
nDI	49.8 (3.1)	4.06 (1.01)
Long leg		
DC	80.9 (11.5)	4.59 (0.81)
DI	81.9 (9.7)	4.01 (0.96)
nDC	81.7 (11.5)	4.50 (0.76)
nDI	79.2 (10.9)	4.14 (0.84)
Statistics (F, p, η_p^2)		
Deformation	92.93, $< .001^{***}$, .912	0.282, .608, .03
Dominance	0.034, .857, .004	0.360, .564, .038
Stimuli	1.174, .307, .115	17.00, $.003^{**}$, .654

41.7 $\pm$ 3.2 cm, meaning the proprioceptive drift corresponded to an average of 74.8% of the virtual leg deformation. This suggests that the perceived foot position aligned more closely with the virtual foot than the actual one. This shift demonstrates the plasticity of body representation in response to lower-limb deformation, which may affect subsequent motor actions due to the interaction between body perception and action [2]. Moreover, this illusion occurred even in the presence of incongruent stimuli. This finding aligns with Kiltani et al. [4], who observed similar proprioceptive drift with incongruent visuotactile stimuli on a lengthened arm. Their findings suggest that drift operates independently of ownership, reflecting distinct mechanisms for ownership and proprioceptive drift — a conclusion supported by our results. A limitation of this study is the lack of investigation into varying leg lengths or extending the indicator's maximal position, which could reduce habituation to the footrest. Future research could address these points.

In a rehabilitation context, injury often leads to altered and reduced proprioceptive input to the affected joint, leading to a higher risk of re-injury. This study offers a novel approach to understanding how altered proprioceptive inputs in the lower limbs can be assessed and potentially rehabilitated using virtual reality.

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